A highly available system can recover from a failure automatically with a short downtime. A fault-tolerant system, in contrast, requires the system to provide its services without interruption in the case of a component failure. We'll show you how to build a fault-tolerant system in chapter 16.

AWS provides building blocks for highly available systems based on EC2 instances. Depending on whether you can distribute a workload across multiple machines, different tools come into play, as described here:

- Building a highly available infrastructure by using groups of isolated data centers, called availability zones, within a region.
- Monitoring the health of virtual machines with CloudWatch and triggering recovery automatically, if needed. This option fits best for workloads that need to run on a single virtual machine.
- Using autoscaling to guarantee a certain number of virtual machines are up and running and replace failed instances automatically. Use this approach when distributing your workload among multiple virtual machines is an option.

You will learn how to automatically replace a failed virtual machine in the first section of this chapter. But what happens if you have a data center outage? You will learn how to recover from that in the second section of this chapter. At the end of the chapter, we will discuss how to analyze disaster recovery requirements and translate them into an AWS architecture. In the following section, you'll learn how to protect a workload that can run on only a single virtual machine at a time from failures.

13.1 Recovering from EC2 instance failure with CloudWatch

AWS does recover failed virtual machines under only some circumstances. For example, AWS will not recover an EC2 instance if a whole rack failed. An easy countermeasure is to automate the recovery of an EC2 instance by creating a CloudWatch alarm. You will learn how to do so in this section.